07
October
2026

WSH ADVISORY

Recent Fatalities Underscore the Need for Safe Concrete Pumping Operations

Concrete pumps are widely used in construction projects to transport and place concrete efficiently at elevated or hard-to-reach areas. However, concrete pumping operations can expose workers to significant hazards, including hose whipping, moving or collapsing booms, pipeline bursts and equipment overturning. These hazards can result in serious injuries or fatalities, particularly when equipment is not properly set up, operated or maintained.

Two fatal accidents involving concrete pump operations occurred in September 2026:

- On 11 September 2026, a worker was struck by a whipping concrete hose after residual high-pressure air was released while clearing a blockage in the hose following concrete pumping operations. He was pronounced dead at the scene.
- On 28 September 2026, a group of workers was carrying out concrete casting works on the 12th storey of a building under construction using a mobile concrete pump. While one of the workers was holding the flexible concreting hose in preparation for casting, concrete suddenly gushed out, causing the hose to whip and strike him. The force of the impact threw the worker against an edge barricade, dislodging it and causing him to fall from the 12th storey to the ground below. The worker sustained fatal injuries and died at the scene.

Figure 1: Overview of the accident scene.

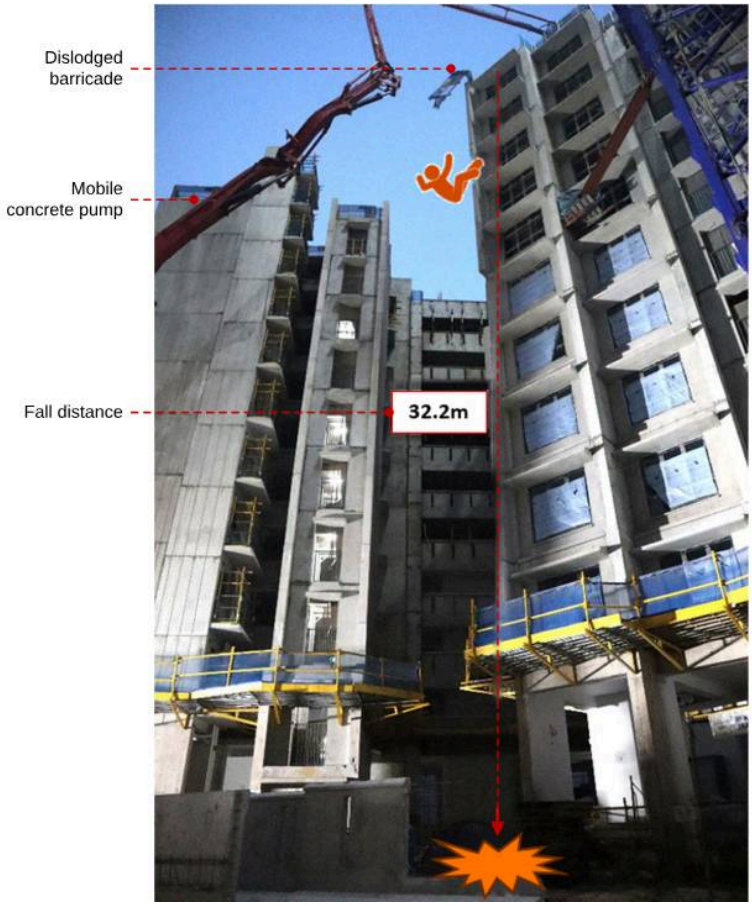


Figure 2: Close-up of work area at 12th storey.



Previous accidents have shown these risks are not new

- On 2 January 2025, a worker was struck by a concrete pump hose after a truck-mounted concrete pump tilted when one of its outriggers punctured the temporary concrete slab supporting it during casting operations.
- On 2 December 2023, a worker was pinned between the chassis of a concrete pump truck and its retracting swivel-type outrigger while the outrigger was being retracted.

These incidents highlight the serious risks associated with concrete pumping operations. Adequate planning, supervision, and control measures are essential to ensure the safe execution of concrete pumping activities.

The WSH Council urges all companies involved in concrete pumping and casting activities to review their risk assessments and work procedures, and to implement effective risk control measures before and throughout the work process.

What companies should do

1. **Plan the operation:** Plan the entire concrete pumping operation, from the arrival of the concrete pump truck on site to its departure after work completion. The plan should cover the ground conditions, pump stability, deployment location and work area, boom and hose operation, and foreseeable risks such as sudden hose movement or whipping caused by blockage.

2. **Establish and follow safe work procedures:** Adhere to pump operating manual and develop task-specific safe work procedures covering all stages of the concrete pumping operation, with particular attention to managing the risks of sudden concrete discharge, hose whipping and working near building edges. Assess factors that may affect pumping pressure and hose movement, such as the concrete mix, pumping height, pipeline configuration, and equipment condition.

3. **Conduct pre-operational checks:** Inspect the concrete pump, pipeline, couplings and flexible hose before use, and ensure they are compatible with the intended operating pressure and in good condition. Replace worn or damaged components and ensure all connections are securely fastened to minimise the risk of hose whipping and uncontrolled concrete discharge.

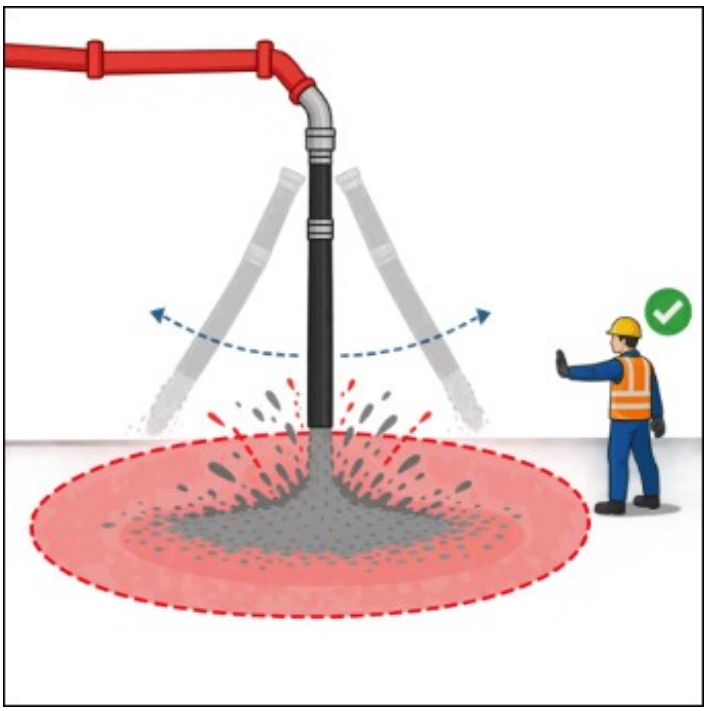
4. **Pump concrete safely:** Work in a safe and clear area. Prime the delivery pipeline before pumping to ensure there are no blockages. Start pumping at a low, controlled rate to minimise pressure surges and risks from sudden hose movement. Gradually increase the pumping rate only after confirming that concrete is flowing smoothly through the system. Stop pumping immediately if there is abnormal pressure, interrupted concrete flow, or unexpected hose movement. Rectify the issue before resuming operation.

> Never attempt to clear a blockage by increasing pump pressure beyond the manufacturer's instructions.

5. **Position workers safely:** Ensure workers handling the flexible hose are positioned away from building edges and are alerted before pumping commences. If working near building edges is unavoidable, consider implementing fall protection measures in addition to edge protection to safeguard workers against falls arising from unexpected hose movement or whipping.

Worker should keep a safe distance away from the discharge hose.

Figure 3: Worker at a safe distance from discharge hose.



(Disclaimer: AI-generated image for illustrative purposes only and does not depict an actual workplace, equipment, vehicle or worker.)

6. **Maintain the pumping system:** Flush and clean the concrete pump, delivery pipeline and flexible hose after each pour to prevent blockages and excessive pressure build-up, which may result in uncontrolled concrete discharge or hose whipping during subsequent operations.

7. **Communicate hazards and emergency actions:** Brief all workers on the hazards associated with pumping activities, including sudden concrete discharge, hose whipping, and fall hazards when working near building edges. Establish clear communication between the pump operator and workers handling the hose, and ensure workers understand the emergency actions to take if abnormal pressure, interrupted concrete flow, or unexpected hose movement is observed (stop pumping immediately and move to a safe location).

For more information

- WSH (Risk Management) Regulations
- WSH (Construction) Regulations 2007
- WSH (Work at Heights) Regulations
- SS 679:2021+C1:2022 Code of Practice for WSH Management System for Construction Worksites
- Code of Practice on WSH Risk Management
- Code of Practice for Working Safely at Heights
- Case Studies for Construction Industry (Case 15)

Under the WSH Act, first-time corporate offenders may be sentenced to the maximum fine of $500,000 whilst individuals can either be sentenced to the maximum fine of $200,000 and/or an imprisonment not exceeding 2 years. Read more on the WSH Act penalties.

Share this WSH Advisory

Connect with Us

Resources to improve your WSH Practices

Events on Workplace Safety and Health

WSHCOUNCIL
Tripartite Alliance for Workplace Safety and Health

We regret that we are unable to reply to emails at this email address. For enquiries or feedback, please email us at contact@wshc.sg. Visit the WSH Council's website now for updates on WSH-related matters, information and events.



This WSH Advisory is available free of charge, may be printed or downloaded onto electronic, optical or similar storage media provided that such activities are for private research, study, or in-house use only. If such material is to be copied or reproduced, users are asked to ensure that the material is reproduced accurately and not in a misleading context, that the status of the material is identified and to acknowledge Workplace Safety and Health Council, Singapore as the source of the material.

Subscribe | Unsubscribe | Privacy Statement